DO NOW — RECALL

Lesson 050 · Queue<T>: FIFO and Service Order

Name: _____ Class: _____ Date: _____

1. What is stack Pop order?

2. What is Peek?

3. What does Enqueue do?

4. An empty queue executes enqueue(A), enqueue(B), enqueue(C), x=dequeue(), enqueue(x), then dequeue(). Trace the final queue from front to rear.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Last-in-first-out.

2. Model answer

Read an end item without removal.

3. Model answer

Adds to rear of queue.

4. Model answer

C, A. The first dequeue stores A in x; A joins the rear, then the final dequeue removes B.

By the end, you can…

• A queue is first-in, first-out (FIFO). Enqueue joins the rear; Dequeue removes from the front.

• Queues are suitable for ordered service requests such as print jobs or breadth-first exploration, if the queueing policy is FIFO.

• Peek inspects the front without removing. An empty Queue<T> throws on Dequeue/Peek unless the code guards against it.

Vocabulary — know these terms

FIFO — First In First Out.

Enqueue — Add at queue rear.

Dequeue — Remove from queue front.

front — Next item to be served.

NEW LEARNING

Read, annotate and explain

Observe

A queue is first-in, first-out (FIFO). Enqueue joins the rear; Dequeue removes from the front.

Explain

Queues are suitable for ordered service requests such as print jobs or breadth-first exploration, if the queueing policy is FIFO.

Apply

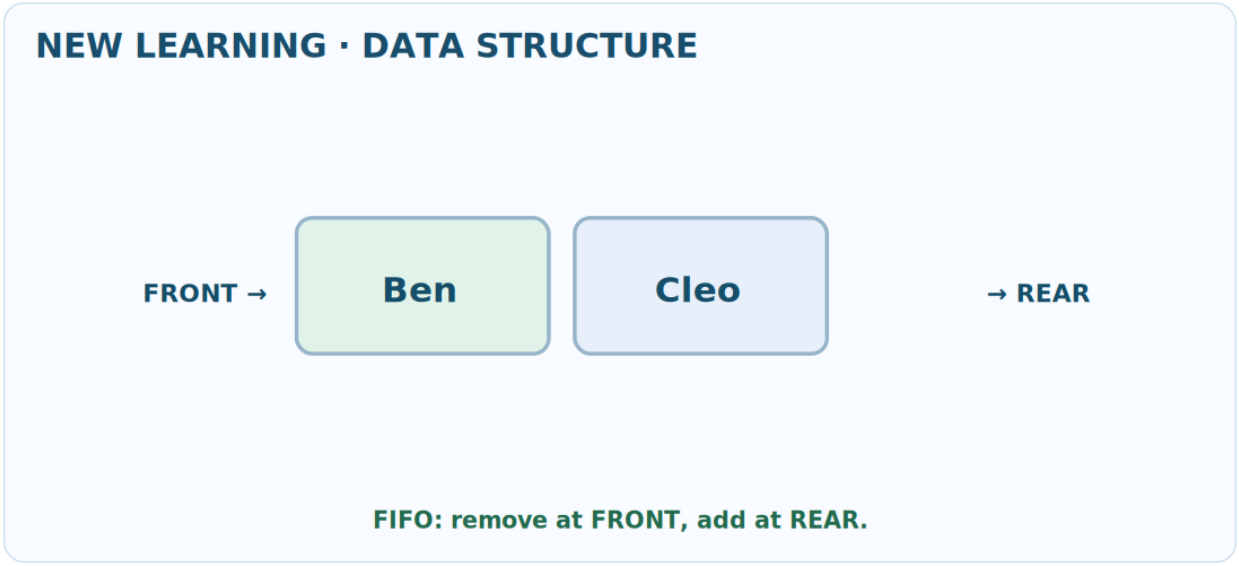
Peek does not remove. Guard Dequeue/Peek with Count>0. Queue<T> handles storage internally; an explicit circular array queue instead tracks front, rear and count and wraps indexes modulo capacity (revisited in Lesson 197).

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



NEW LEARNING · DATA STRUCTURE: Ben, Cleo

I DO — WORKED EXAMPLE

Follow the worked method

Trace a ticket queue with Jo then Kai; dequeue one.

Trace and justify

Jo is served first; Kai remains at the front.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace a ticket queue with Jo then Kai; dequeue one.

Jo is served first; Kai remains at the front.



IDO · WORKED STATE: Kai

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A different queue receives jobs 4, 7, 2. Dequeue twice, enqueue 9. What remains?
Source code / input:

```
var q=new Queue<int>();
q.Enqueue(4);q.Enqueue(7);q.Enqueue(2);
q.Dequeue();q.Dequeue();q.Enqueue(9);
```

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Queue A,B,C; Dequeue once, then Peek. State removed and front value. [6]

Question 2

Write C# code to enqueue 2,4,6 into Queue<int>, then safely dequeue one. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Compare Stack<T> and Queue<T> for DFS versus BFS on a graph, explaining removal order. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A queue's Dequeue removes the newest item first.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

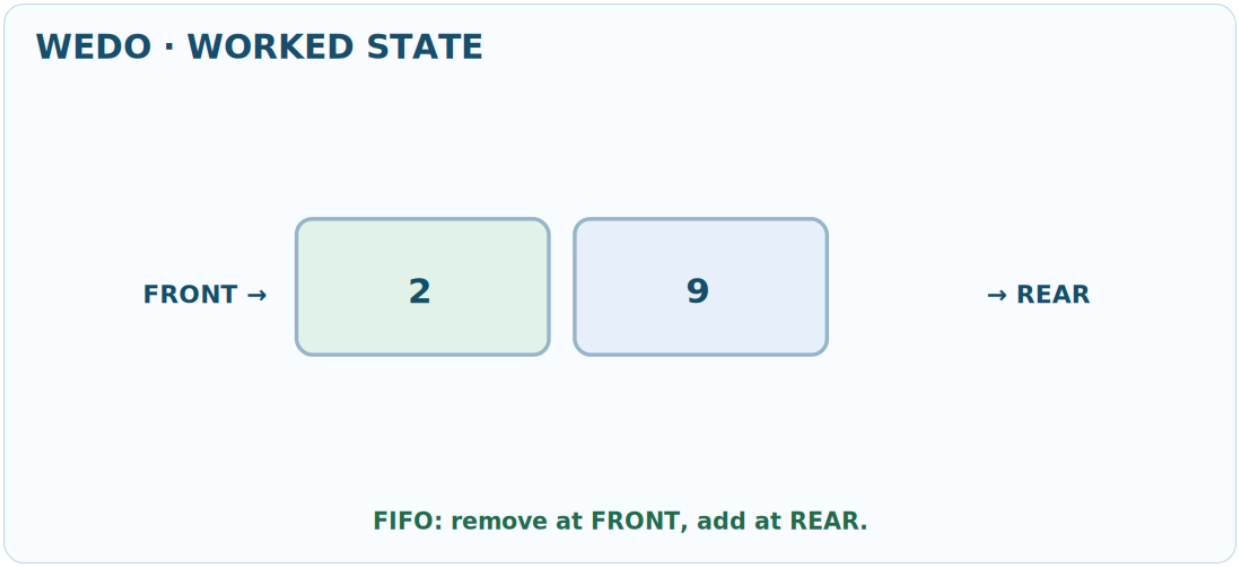
Trace and justify
Dequeue returns 4 then 7. Remaining front-to-back [2,9].

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A different queue receives jobs 4, 7, 2. Dequeue twice, enqueue 9. What remains?

Dequeue returns 4 then 7. Remaining front-to-back [2,9].



WEDO · WORKED STATE: 2, 9